

Chuyao Fu (Troy)

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EDUCATION

Southern University of Science and Technology (SUSTech)

B.Eng. in Electronic Information Engineering

Sep. 2023 – Jun. 2027

GPA: 3.90/4.00 Rank: 3/46

- **Selected Coursework:** Data Structures and Algorithm Analysis, Probability and Statistics, Machine Learning, Robotic Motion and Control, Digital Signal Processing, Signals and Systems, Analog Circuits, Digital Circuits

RESEARCH EXPERIENCE

HMI Lab, School of Computer Science, Peking University

Undergraduate Researcher

Jun. 2025 – Present

Supervisor: Prof. Shanghang Zhang

- Leading **Token-World**, which studies direct action-conditioned simulation for VLA policies in compact visual-token space without intermediate RGB reconstruction.
- Leading **VA-CLIP**, a reference-free verifier that learns action-video compatibility to evaluate and select embodied world-model rollouts.
- Contributed to **LJM**, an action-conditioned world model that predicts robot-environment interaction latents before generating future videos.
- First author on both under-review manuscripts; responsible for problem formulation, method design, experiments, and manuscript preparation.

Robotics and Computer Vision Lab, SUSTech

Undergraduate Researcher

Sep. 2024 – Present

Supervisors: Prof. Hong Zhang and Prof. Max Q.-H. Meng

- Proposed **ProDrive**, a proactive autonomous-driving framework that jointly trains a planner and a BEV world model through ego-environment co-evolution; first author and lead researcher.
- Co-led **AxisField-GS**, a feed-forward articulation-recovery framework that predicts dense Gaussian-level motion fields and recovers revolute joint axes from single-state semantic 3DGS assets; co-first author.

SELECTED PUBLICATIONS AND MANUSCRIPTS

ProDrive: Proactive Planning for Autonomous Driving via Ego-Environment Co-Evolution

Chuyao Fu, Shengzhe Gan, Zhuoli Ouyang, Yuhan Rui, Xiaowei Chi, Sirui Han, Jiankun Wang, Hong Zhang

CVPR 2026 GigaBrain Challenge Workshop

Learning Action-Video Compatibility as a Reference-Free Verifier for Embodied World Models

Chuyao Fu, Xiaojie Zhang, Xiaowei Chi, Rongyu Zhang, Yuhan Rui, Jiajun Cao, Yike Guo, Sirui Han, Shanghang Zhang

Under review

Token-World: Direct World Model Simulator for Vision-Language-Action Models

Chuyao Fu, Xiaowei Chi, Haoran Li, Yuhan Rui, Zezhong Qian, Yu-Kai Wang, Xiaojie Zhang, Yunfan Lou, Hongyang Cheng, Yike Guo, Sirui Han, Shanghang Zhang

Under review

AxisField-GS: Feed-Forward Articulation Recovery from Single-State Semantic Gaussian Assets

Shengzhe Gan*, Chuyao Fu*, Yugong Wang, Guangcheng Chen, Hong Zhang. *Equal contribution

Under review

Mask World Model: Predicting What Matters for Robust Robot Policy Learning

Yunfan Lou, Xiaowei Chi, Xiaojie Zhang, Zezhong Qian, Chengxuan Li, Rongyu Zhang, Yaoxu Lyu, Guoyu Song, Chuyao Fu, Haoxuan Xu, Pengwei Wang, Shanghang Zhang

ICML 2026

EchoArena: Mutual Benchmarking of World Models and VLA Policies

Yu-Kai Wang, Kevin Zhang, Xiaowei Chi, Tianxing Chen, Siqiao Huang, Chuyao Fu, Tiecheng Guo, Peidong Jia, Yan Qin, Kuangzhi Ge, Siyuan Qian, Weishi Mi, Zezhong Qian, Jiajun Li, Qingpo Wuwu, Shanghang Zhang, Jian Tang

CVPR 2026 GigaBrain Challenge Workshop

FORCE: Efficient VLA Reinforcement Fine-Tuning via Value-Calibrated Warm-up and Self-Distillation

Shuyi Zhang, Yunfan Lou, Hongyang Cheng, Yichen Guo, Chuyao Fu, Yaoxu Lyu, Xiaojie Zhang, Haoran Li, Pengwei Wang, Zhongyuan Wang, Shanghang Zhang

arXiv preprint, 2026

LJM: A Latent Joint-conditional Model for Action-Conditioned Embodied World Model

Kuangzhi Ge, Yu-Kai Wang, Yuhan Rui, Xiaojie Zhang, Hao Chen, Jiaming Liu, Chuyao Fu, Zhi Yang Chen, Tianze Zhou, Zhuoqun Wu, Athend Zhuoming Zhong, Chak-Wing Mak, Kevin Zhang, Xiaowei Chi, Shanghang Zhang, Sirui Han

Under review

INDUSTRY EXPERIENCE

Muka Robotics

Research Intern

Jul. 2026 – Present

Beijing

- Developing representation learning techniques and data cleaning pipeline for foundational world-action-model pretraining.
- Built a physical world model demo in which a robot observes, predicts, and plays tabletop curling against a human.

XtalPi

Research Intern, Future Chemistry Department

Mar. 2026 – Apr. 2026

Shenzhen

- Trained and deployed VLA policies, including LingBot-VLA, π_0 , and $\pi_{0.5}$, for robot manipulation in laboratory-automation settings.
- Integrated perception, policy inference, and control on real robot platforms; conducted iterative evaluation and ablations to diagnose deployment failures.

Beijing Academy of Artificial Intelligence

Research Intern, Embodied Multimodal Large Models Research Center

Jun. 2025 – Sep. 2025

Beijing

- Processed large-scale embodied datasets, including AgiBot-World and DROID, for VLA pretraining; implemented action normalization, tokenization, and structured training-data conversion.
- Reproduced representative VLA reinforcement-learning and robot-policy post-training frameworks; contributed to **FORCE** and the cross-embodiment **RoboBrain-X0** project.

SELECTED PROJECTS

Awesome Representation for World Models

Open-Source Literature Curation

Aug. 2026 – Present

Project Lead

- Currently lead a collaborative survey of representations for world models, coordinating lab members to curate literature around state and transition representations.
- Designed a Cloudflare-based webpage workflow spanning literature review, taxonomy-based classification, and pull-request review for multi-contributor maintenance.

RoboBrain-X0: Cross-Embodiment Vision-Language-Action Model

Open-Source VLA Model

Jun. 2025 – Sep. 2025

Contributor

- Contributed to **RoboBrain-X0**, a unified VLA model for cross-embodiment action generation that supports heterogeneous robot platforms through a shared action-token representation.
- Focused on the data pipeline, processing AgiBot-World and DROID with action normalization, tokenization, and structured conversion for large-scale VLA pretraining.

LEADERSHIP AND SERVICE

Class Leader, Class 2303

Class Committee

2024 – 2025

- Led class operations and activities, including event planning, logistics coordination, routine administration, and award application materials; the class received honors including *Excellent Undergraduate Academic Style*, *Best Sports Team*, and *Best Popularity Class*.

House Division Lead, Street Dance Club

Student Organization

2024 – 2025

- Led the House division by coordinating weekly training with instructors and organizing two campus street dance performances, including rehearsal planning and member coordination.

HONORS AND AWARDS

- First Prize, Outstanding Student Award, SUSTech (top 5%; RMB 5,000 scholarship), Sep. 2025
- Third Prize, Outstanding Student Award, SUSTech (top 25%; RMB 1,000 scholarship), Sep. 2024
- First-Class Admission Scholarship, SUSTech (top 10%; RMB 50,000 scholarship), Sep. 2023
- Gold Medal, Chinese Chemistry Olympiad, Chinese Chemical Society (top 150 nationwide), Nov. 2022

SKILLS

- **Programming and ML:** Python, PyTorch, Linux, Git
- **Robotics:** ROS, real-robot VLA deployment, camera calibration, 3D vision, and point-cloud processing
- **Languages:** Chinese (native); English (TOEFL iBT 100: R25, L25, S23, W27)